Snowshoe hare and lynx population cycles

In Canada's boreal forests, the favored prey of lynx are snowshoe hares. Charles Elton examined the relationship between the populations of these two species, using data covering the period 1845–1925. When hares are numerous, lynx hunt little else. After their population reaches its peak density, the hares struggle to find enough plant food. Some starve, while others are weakened and are more easily caught by predators, including lynx, which feed very well for a time. When hare numbers continue to fall, this affects the lynx. They are forced to hunt less nutritious prey, such as mice and grouse.

As they struggle to find enough to eat, lynx produce smaller litters or even stop breeding altogether. Some starve to death. A decline in the lynx population sets in one or two years after the hare population has bottomed out, a cycle that repeats every eight to eleven years.